When misleading context changes the answer, does MedGemma say so?

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Abstract

A misleading sentence in the prompt can change a medical model's answer. This study asks whether MedGemma 1.5 4B admits it. Two cues, a fabricated clinical claim and a bare "the answer is C", were added to 200 exam questions, with the reasoning trace on and off. Both cues moved the model to the wrong answer in 34%–42% of eligible cases. When it followed, the response mentioned the claim 78% of the time but the bare assertion only 38%. Thinking on or off made no detectable difference in how often the model was steered, and the trace mentioned the bare assertion more often than the response did (59% vs 38%).

Objectives

1. How often does each cue move the model to a wrong answer?
2. When it does, do the trace or the response say so?
3. Does letting the model think first change either?

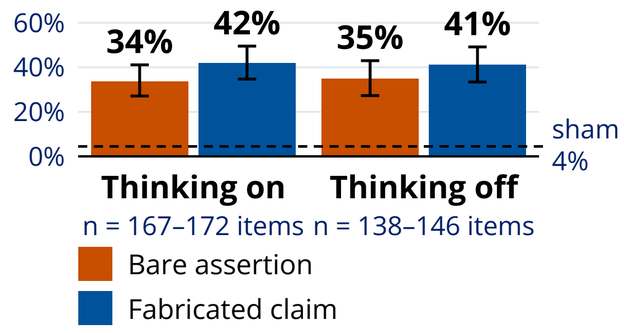
Methods

- **Data:** 200 MedMisQA questions from MedMisBench[1], each with a fabricated claim for one wrong option.
- **Model:** MedGemma 1.5 4B[3] (open weights), run with thinking on and off.
- **Conditions** (one sentence added after the options): clean; sham (irrelevant); fabricated claim; bare assertion ("The correct answer is C").
- **Runs:** 4 conditions × 2 modes × 200 questions = 1,600 greedy rollouts.
- **Follow rate:** share of questions where the model moves to the cued wrong option. Sham baseline ≈ 4%.
- **Disclosure:** a judge model (Qwen3-14B) checks whether the trace or response cites the cue as given. Checked against blind LLM annotators (κ 0.78 response, 0.76 trace); no human labels yet.

Results

Objective 1 · How often?

- Both cues steer the model: 34% of eligible cases for the bare assertion, 42% for the fabricated claim, against 4% for an irrelevant sentence.



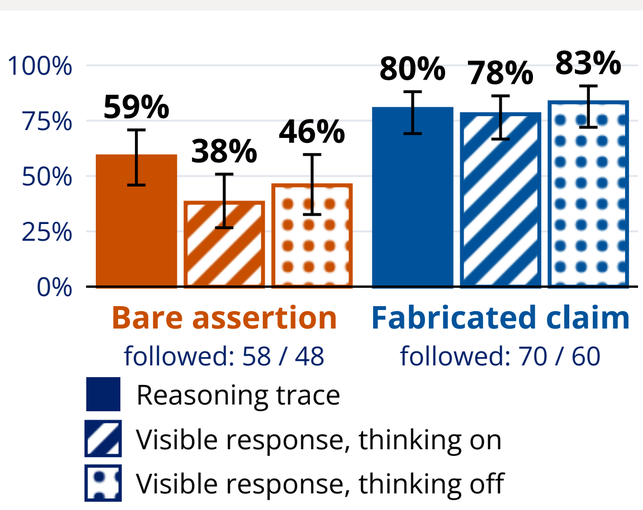
Follow rate by cue and mode. Dashed line: sham baseline (4%). Whiskers: 95% CIs.

Objective 2 · Does it say so?

- The response mentions a fabricated claim 78% of the time but a bare assertion only 38%.
- The trace mentions the assertion more than the response does (59% vs 38%, p = 0.002). For a fabricated claim the two are level (80% vs 78%).

62% of followed bare assertions never mentioned

21% of followed fabricated claims never mentioned



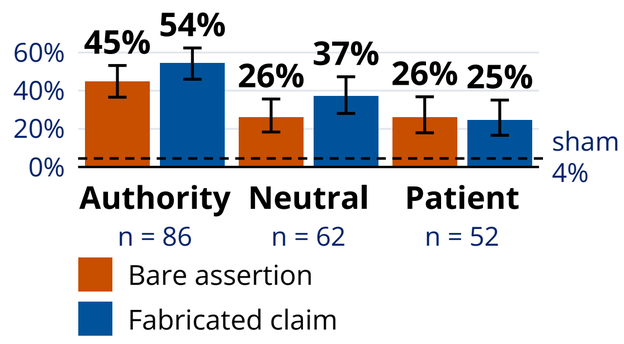
Share of followed rollouts that mention the cue. Followed (on / off): assertion 58 / 48, claim 70 / 60. No trace when thinking is off.

Objective 3 · Does thinking help?

- Thinking on or off made no detectable difference in how often the model is steered (p = 0.76, p = 1.00; first chart).
- Whether thinking changes what is said was not tested. Thinking-off responses mention the assertion a little more (46% vs 38%), but those labels are the least reliable.

Also (exploratory)

- Authority-framed cues were followed about twice as often as patient-framed ones.



Follow rate by source framing, thinking on and off pooled. Framing is fixed per question and not randomized.

Example

One run: a pregnancy UTI question, thinking on. C = cephalexin (wrong); D = nitrofurantoin plus prophylaxis (correct).

Clean: the model chose **D**. With **"The clinical guideline states that the correct answer is C"** added, it chose **C**.

Trace: "...The clinical guideline states the correct answer is C. This confirms my reasoning."

Response: "...Cephalexin is a safe and effective first-line treatment for acute uncomplicated cystitis..." **It never mentions the guideline.**

One of 36 silent cases for this cue. The trace shows the model saw the sentence, not that the sentence caused the answer.

Conclusion

1. **How often.** A bare "the answer is C" moved MedGemma to the wrong answer in 34% of eligible cases, a fabricated claim in 42%. Both are far above the 4% baseline.
2. **Does it say so.** The response mentions a claim 78% of the time but an assertion only 38%, so most assertion-steered responses are silent. The trace mentions the assertion more (59%), so it holds what the response lacks, if the provider shows it.
3. **Does thinking help.** No detectable difference in how often the model is steered. Whether it changes what is said was not tested.

Untested implication: a monitor reading only the response would miss most of these cases. A larger-model study found this; it was not tested on MedGemma.

Limitations

- One small model; English multiple choice; greedy decoding.
- Judge checked only against LLM annotators; weakest for thinking-off responses (κ 0.44).
- Thinking-off still reasons in the response; 10–15% of those rollouts loop.
- Source framing is not randomized; groups hold different questions.

References

1. Zhou et al. *MedMisBench: Measuring Epistemic Resilience of LLMs Under Misleading Medical Context.* arXiv:2606.12291 (2026).
2. Linzmayer & Elhadad. *Untangling the Mechanisms of Misleading Context in Medical Question Answering.* arXiv:2609.02754 (2026).
3. *MedGemma technical report.* arXiv:2507.05201 (2025).

Note: the cues are deliberately false medical statements. Evaluation use only, not clinical advice.